Metaboenrich step 5 (default): Ohm's-law current flow on the metabolite–reaction network

Metaboenrich methods note

Code reference: `metaboenrich/flow.py`, commit `0f6c135` (branch `directed-rw`)

Abstract

Metaboenrich ranks genes, reactions and pathways from a differential metabolite abundance (DAA) table. Its default scoring step treats the metabolic network as an electrical circuit. Every metabolite that went up is a current source and every metabolite that went down is a sink. For each source–sink pair, one unit of current is injected and Kirchhoff's and Ohm's laws give the current on every edge. Reactions that carry a large share of the current, or that are the only route for some pair, rank high. This note defines the method exactly as implemented, shows how it is solved efficiently, and lists its properties and limits. A companion note describes the experimental alternative, a directed absorbing random walk.

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1 Where this step sits in the pipeline

The main run (`python -m metaboenrich`) has seven steps:

1. load Human-GEM v2.0.1 (`gem.py`);
2. decide which metabolites are currency (`network.py`);
3. map the DAA table onto model metabolites (`daa.py`);
4. build the metabolite–reaction graph (`network.py`);
5. **solve the current flow** (`flow.py`, this note);
6. score and rank reactions, genes and pathways (`scoring.py`);
7. write the outputs (`__main__.py`, `cytoscape.py`).

Step 5 takes the graph from step 4 and the source and target lists from step 3, and returns per-node and per-edge arrays (a `FlowResult`) that step 6 turns into ranked tables. It corresponds to protocol §1.4 (current flow) and §2.2 (directed pair filter).

2 The network as a circuit

2.1 Graph

The graph $G = (V, E)$ is bipartite. $V = M \cup R$ holds one node per metabolite (compartments merged) and one node per reaction. An edge joins a metabolite to a reaction it takes part in, as substrate or product. Currency metabolites (H^+, H_2O, phosphate, cofactor exchange pairs such as ATP/ADP, ...) are removed before the graph is built, so they cannot act as shortcuts. On Human-GEM the locked network has 3,958 metabolites, 6,377 reactions and 16,356 edges.

Every edge has unit conductance (resistance $1\,\Omega$). Edges are *undirected*: reaction directions are not used in the electrical problem.

2.2 Incidence matrix and Laplacian

Number the edges $e = 1, \ldots, m$ and the nodes $v = 1, \ldots, n$. The signed incidence matrix $B \in \mathbb{R}^{m \times n}$ has, for edge $e = (\mu, \rho)$ joining metabolite μ and reaction ρ,

$$B_{e\mu} = +1, \qquad B_{e\rho} = -1, \qquad B_{ev} = 0 \text{ otherwise.}$$

The orientation is a bookkeeping convention: a positive edge current means current flows from the metabolite into the reaction. The weighted graph Laplacian with unit conductances is

$$L = B^\top B \in \mathbb{R}^{n \times n}.$$

L is symmetric and positive semi-definite. Its null space is spanned by the indicator vectors of the connected components, so L is singular.

3 One source–target pair

3.1 Kirchhoff and Ohm

For a source s ($\log_2\text{FC} > 0$) and a target t ($\log_2\text{FC} < 0$) in the same connected component, inject one ampere at s and withdraw it at t:

$$L\,x^{st} = b^{st}, \qquad b^{st} = e_s - e_t, \tag{1}$$

where $x^{st} \in \mathbb{R}^n$ are the node potentials and e_v is the v-th unit vector. Ohm's law gives the edge currents

$$I^{st} = B\, x^{st} \in \mathbb{R}^m, \qquad I_e^{st} = x_\mu^{st} - x_\rho^{st} \text{ for } e = (\mu, \rho). \tag{2}$$

Kirchhoff's current law holds by construction, $B^\top I^{st} = b^{st}$: one unit leaves s, one unit arrives at t, and current is conserved at every other node.

3.2 Throughput

The throughput of node v for the pair (s, t) is the current passing through it,

$$T^{st}(v) = \tfrac{1}{2} \sum_{e \ni v} \left| I_e^{st} \right| \quad (v \neq s, t), \qquad T^{st}(s) = T^{st}(t) = 0. \tag{3}$$

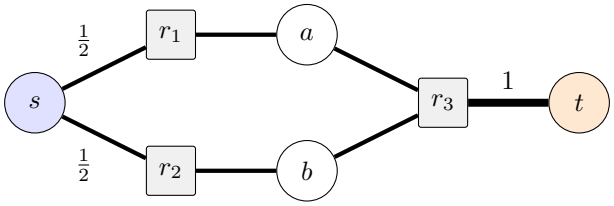
The factor $\frac{1}{2}$ counts current once: what enters an intermediate node also leaves it. $T^{st}(v) \in [0, 1]$, and $T^{st}(v) = 1$ means every path from s to t passes through v (a bottleneck). The endpoints are set to zero because they trivially carry the whole unit. This is the current-flow betweenness of Newman (2005) and Brandes and Fleischer (2005), restricted to the measured pairs.

3.3 Effective resistance

The potential drop between the endpoints is the effective resistance,

$$R_{\text{eff}}(s, t) = x_s^{st} - x_t^{st},$$

small when s and t are joined by many short, parallel routes. For each source or target the output reports the mean effective conductance $1/R_{\text{eff}}$ over its pairs (`eff_conductance`) and the mean largest current on any one of its own edges (`max_edge_share`), which shows whether its current leaves through a single reaction.



$T(r_1) = T(r_2) = T(a) = T(b) = \frac{1}{2}, \quad T(r_3) = 1$ (bottleneck)

Figure 1: Current splits over parallel routes and recombines. Reaction r_3 is the only way into t, so it carries the full unit: bottleneck = 1.

4 All pairs at once: grounding and one factorisation

Solving (1) separately for all $|S| \times |T|$ pairs, or forming the pseudoinverse L^+ (dense, $O(n^3)$), would be wasteful. The implementation uses linearity instead.

Grounding. In each connected component C, fix the potential of its first node g_C to zero. Deleting that row and column gives a non-singular matrix L_C. Grounding shifts all potentials by a constant, which leaves edge currents and effective resistances unchanged.

Unit solves. For each source or target v in C, solve once

$$L_C\, y^v = e_v, \tag{4}$$

the potentials when one unit enters at v and leaves at the ground. Each component is LU-factorised once (`scipy.sparse.linalg.splu`) and reused for all its right-hand sides.

Pairs by subtraction. For any pair in C,

$$x^{st} = y^s - y^t,$$

because the right-hand sides subtract to $e_s - e_t$ and the ground terms cancel. So $|S| + |T|$ solves cover all $|S|\,|T|$ pairs. Pairs are then processed in blocks of 128: $I = BX$ and $T = \frac{1}{2}|B|^\top |I|$ are sparse–dense products.

5 Combining pairs into scores

Let P be the retained pairs and w_{st} the pair weights. By default every pair has $w_{st} = 1$ (one ampere per pair, as in the protocol). With `-pair-weight fc`, $w_{st} = \frac{1}{2}(\varphi_s + \varphi_t)$ with $\varphi_v = |\log_2 \mathrm{FC}_v|$. With $W = \sum_P w_{st}$:

$$\text{accumulated current} \quad A(v) = \textstyle\sum_{(s,t)\in P} w_{st}\, T^{st}(v), \tag{5}$$

$$\text{conductivity} \quad c(v) = A(v)/W, \tag{6}$$

$$\text{bottleneck} \quad \beta(v) = \max_{(s,t)\in P} T^{st}(v), \tag{7}$$

$$\text{bottleneck fraction} \quad f_\tau(v) = \tfrac{1}{W} \textstyle\sum_P w_{st}\, \mathbf{1}\big[T^{st}(v) \geq \tau\big], \quad \tau = 0.5, \tag{8}$$

$$\text{edge current} \quad \bar{I}_e = \tfrac{1}{W} \textstyle\sum_P w_{st}\, |I_e^{st}|, \qquad I_e^{\max} = \max_P |I_e^{st}|. \tag{9}$$

Conductivity is the share of the total current that passes through v; the bottleneck is the largest share for any single pair.

5.1 Leave-one-out scores

A reaction can score highly only because one measured metabolite has a single way out. The leave-one-out scores ask whether the score survives dropping any one metabolite. For a source k, let $A_k(v)$ and W_k be the sums over the pairs that use k, and $\beta_k(v)$ the maximum over those pairs. A metabolite is *droppable* if $0 < W_k < W$ (dropping it leaves some pairs). Then

$$c^{\mathrm{loo}}(v) = \min_{k \text{ droppable}} \frac{A(v) - A_k(v)}{W - W_k},$$

taken over droppable sources and targets alike, and clipped at 0. The leave-one-out bottleneck is the second-largest of the per-metabolite maxima $\beta_k(v)$, the worst case after dropping the metabolite with the largest one. If nothing is droppable (one source and one target), the raw values are used. Raw scores are the default (`-scoring raw`); `-scoring loo` ranks on these columns instead.

6 Variants

Pair exclusion. Pairs in different connected components cannot carry current and are written to `excluded_pairs.tsv` ("disconnected").

Directed mode (`-mode directed`, protocol §2.2). Pairs with no directed path $s \to t$ along reaction directions are also excluded. This is *only* a filter: the remaining pairs are solved on the same undirected Laplacian, so current can still run against a reaction's direction. Directed current (protocol §2.3) is not implemented; the directed absorbing random walk is the experimental alternative.

One-sided data (`-one-sided auto`). If only increases (or only decreases) are significant, the measured but unchanged metabolites form a ground set $\mathcal{G}$. Each source injects one unit withdrawn evenly across the ground nodes in its component,

$$b = e_s - \tfrac{1}{|\mathcal{G}_C|} \textstyle\sum_{g \in \mathcal{G}_C} e_g,$$

and symmetrically when only targets exist. Ground nodes are sinks and are not scored.

7 From node scores to ranked genes and pathways

Step 6 is shared with the random walk.

- Reactions with $c > 0$ get a final score: the mean of the percentile ranks of conductivity and bottleneck (raw or leave-one-out).
- A gene takes the best final score among the reactions it catalyses (gene–protein–reaction and/or logic is ignored).
- Pathways (Human-GEM subsystems) are tested for over-representation among the top 100 reactions with a hypergeometric test against all network reactions, with Benjamini–Hochberg correction.
- The Cytoscape subnetwork keeps edges with $I_e^{\max} \geq 0.1$.

8 Properties and limits

- **Direction-blind.** The circuit is undirected, so current can run through an irreversible reaction backwards.
- **Co-substrate hops.** Current can enter a reaction through one substrate and leave through another (for $A + B \to C$, the route $A \to r \to B$), a step no enzyme performs. Currency removal limits the worst cases but does not prevent this in general.
- **Every pair counts equally** by default; fold change enters only through `-pair-weight fc`, and only as a pair weight, not in $Lx = b$.
- **Bounded and interpretable.** Throughput lies in $[0, 1]$; a value of 1 is a cut vertex for that pair.
- **Global.** Current spreads over every parallel route, so long alternative routes still receive some current. This makes the method robust to single missing reactions, but dilutes the signal in dense regions.
- **Exact and fast.** One sparse LU per component; the full run on Human-GEM takes a few seconds.

Remark (Link to random walks)**.** *On an undirected graph, the unit current from s to t equals the expected net number of times a simple random walker started at s and stopped at t crosses each edge (Newman, 2005). Current flow is therefore already a random-walk method, on an undirected graph, one pair at a time, with every step equally likely. The directed absorbing random walk changes exactly those three things.*

9 Benchmark

Inborn errors of metabolism (plasma z-scores from Thistlethwaite et al. 2020 and Miller et al. 2015): rank of the known causal gene among the 2,302 network genes, raw scoring; lower is better. Source: `docs/decisions.md`, D17.

Level	Measure	Current flow	Absorbing walk
Group (20 primary disorders)	median rank	88.5	48
	in top 100	11	13
	in top 10	3	5
Patient (159 primary patients)	median rank	134	67
	in top 100	47%	55%
	in top 10	5%	28%

Current flow remains the default: the blind predictions on published datasets (D14) were made with it, and the walk's gains come from one benchmark.

10 Implementation map

Code (`metaboenrich/`)	What it computes
`network.Graph.incidence()`	B; $L = B^\top B$
`network.Graph.components()`	connected components (grounding, pair filter)
`flow.select_pairs()`	retained and excluded pairs; directed filter
`flow._potentials()`	grounded LU per component, unit solves (4)
`flow.current_flow()`	$I = BX$, throughput (3), A, c, β, f_τ, edge currents, R_{eff}
`flow._leave_one_out()`	c^{loo}, β^{loo}
`scoring.reaction_table()`	final score from percentile ranks
`scoring.gene_table()`, `pathway_table()`	gene ranking; pathway enrichment

References

- Newman, M. E. J. (2005). A measure of betweenness centrality based on random walks. *Social Networks* 27, 39–54.
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- Doyle, P. G. and Snell, J. L. (1984). *Random Walks and Electric Networks.* Mathematical Association of America.
- Metaboenrich protocol, §1.4 and §2.2 (`Metaboenrich-Protocol.pdf`).